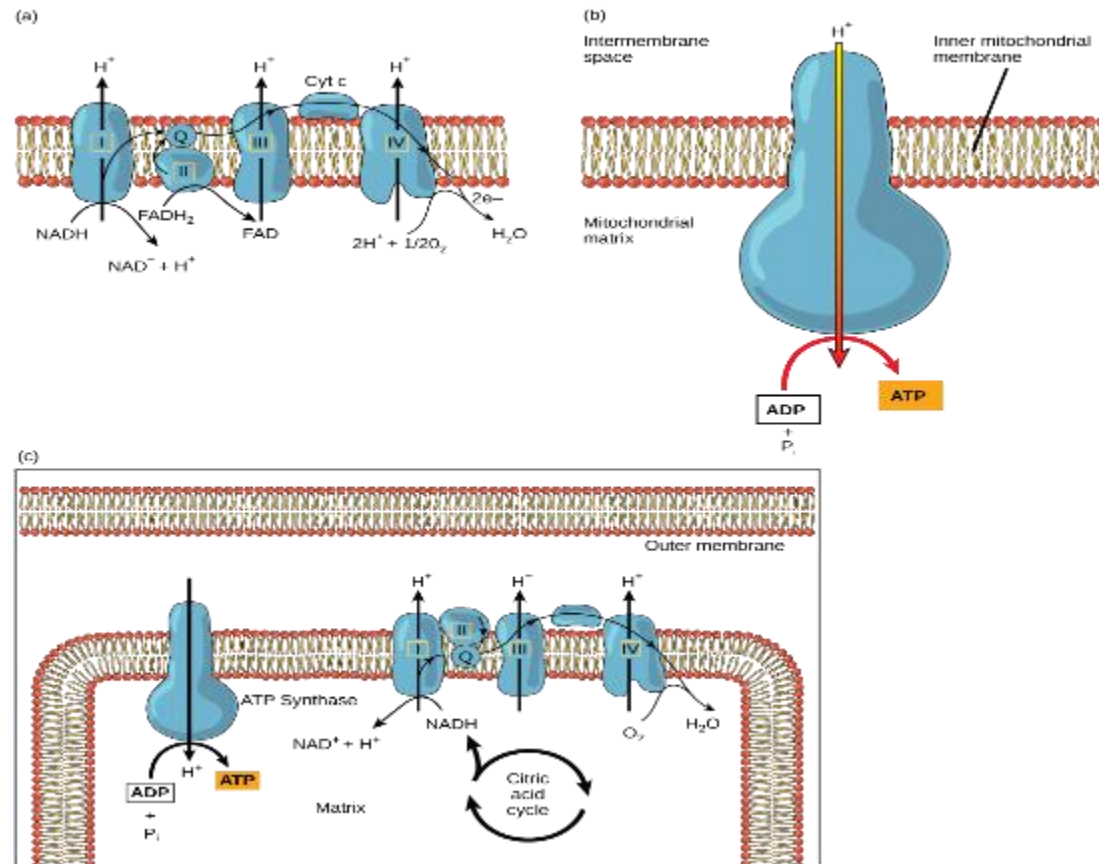


# Glycolysis, Pyruvate processing & citric cycle

- Overall:
  - Follow the carbon: 6-carbon atoms in 1 glucose molecule becomes 6 CO<sub>2</sub> molecules (exhaled out during respiration)
- Energy-containing molecules are made:
  - ATP
  - NADH, FADH<sub>2</sub>
    - These will provide energy for the next step (!)

# oxidative phosphorylation

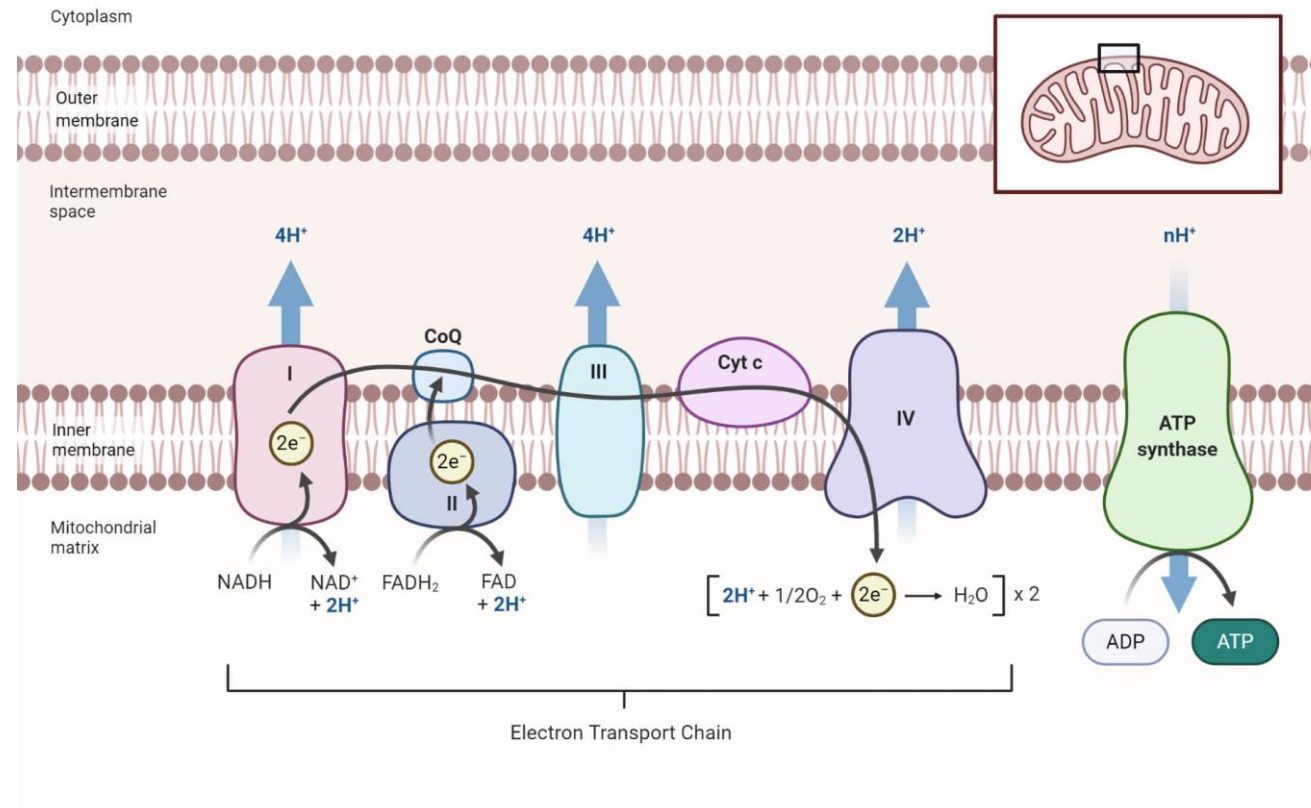
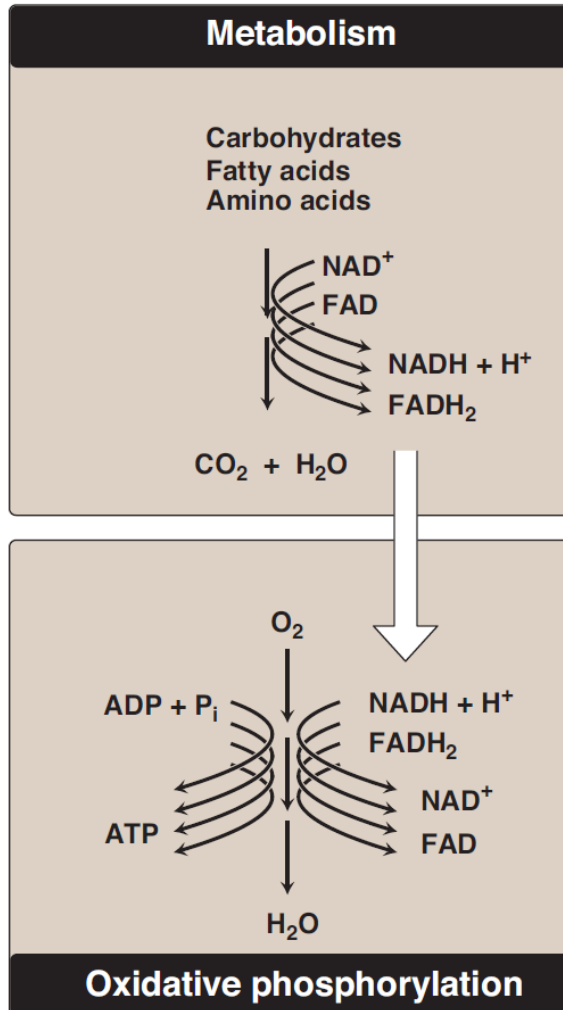


- (a) The electron transport chain is a set of molecules that supports a series of oxidation-reduction reactions.
- (b) ATP synthase is a complex, molecular machine that uses an  $H^+$  gradient to regenerate ATP from ADP.
- (c) Chemiosmosis relies on the potential energy provided by the  $H^+$  gradient across the membrane.

# Oxidative phosphorylation

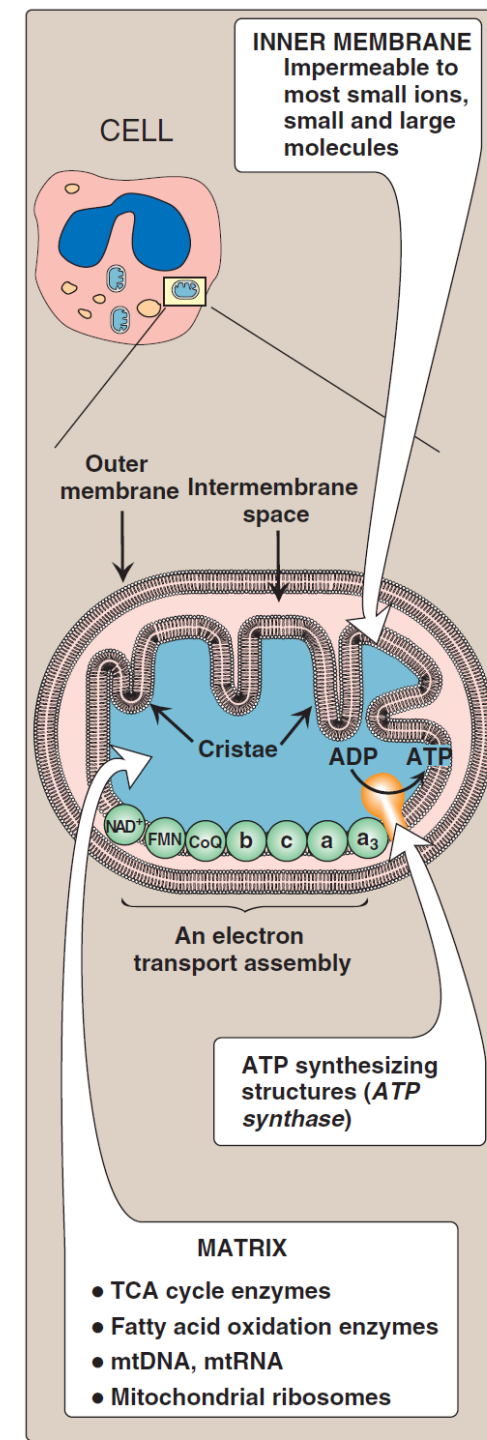
- Oxidation:
  - Losing electrons
  - NADH & FADH<sub>2</sub> will donate (this is oxidation) electrons to the ETC
  - ETC = electron transport chain
    - Made of a chain of proteins
    - Will use the energy of the electrons to create a H<sup>+</sup> gradient
      - Moves H<sup>+</sup> across the membrane resulting in higher concentration in the intermembrane space than the matrix
    - Will transport the electrons to O<sub>2</sub> making H<sub>2</sub>O
- Phosphorylation:
  - Process of adding phosphate to a molecule
  - H<sup>+</sup> will passively move down the gradient through ATP-synthase (from intermembrane space back to matrix)
  - This makes ATP-synthase create ATP from ADP + 1 P
    - 90% of ATP (energy) is made this way
  - Chemiosmosis = coupling of chemical reaction of making ATP to passive transport of H<sup>+</sup> down their concentration gradient

# ELECTRON TRANSPORT CHAIN



[Bing Videos: ETC](#)

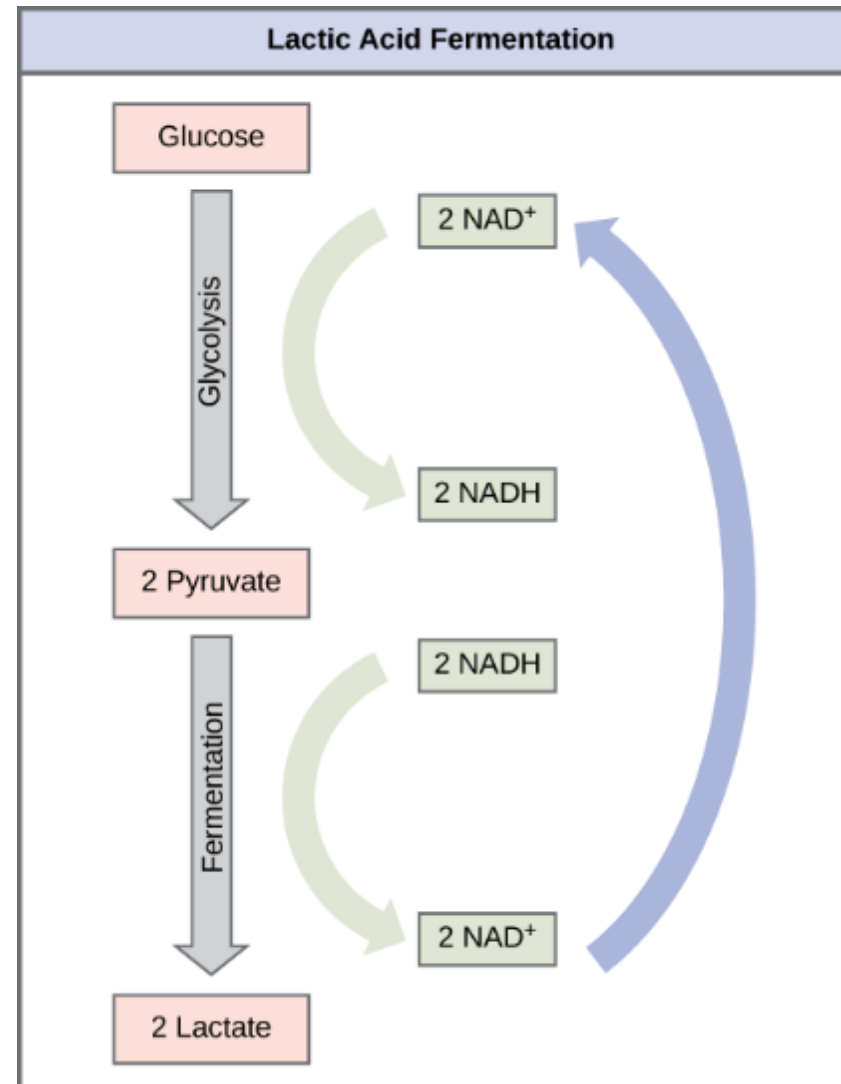
[Bing Videos: ATP synthase](#)



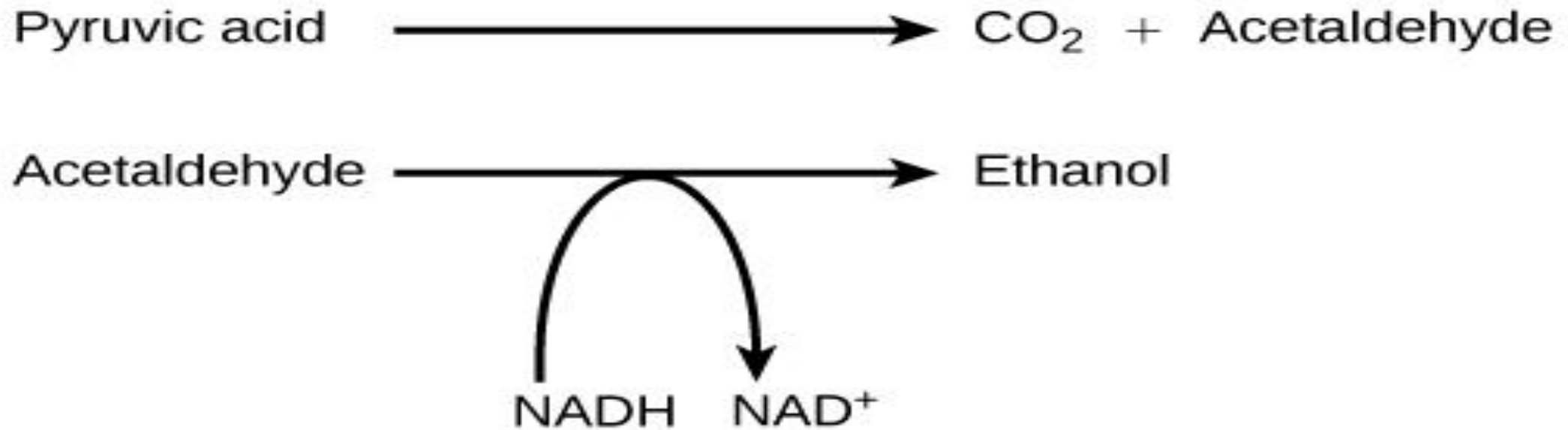
# Fermentation & energy

- Fermentation – obtaining energy without the need of O<sub>2</sub>
  - Makes less ATP than cellular respiration
  - Sugar goes only through glycolysis (!)
- Two types:
  - Lactic acid fermentation – in muscles
  - Alcohol fermentation – in yeast – makes alcohol and gas

- Lactic acid fermentation is common in muscles that have become exhausted by use.



# Figure 4.17



- The reaction resulting in alcohol fermentation is shown.



- Fermentation of grape juice to make wine produces  $\text{CO}_2$  as a byproduct. Fermentation tanks have valves so that pressure inside the tanks can be released.

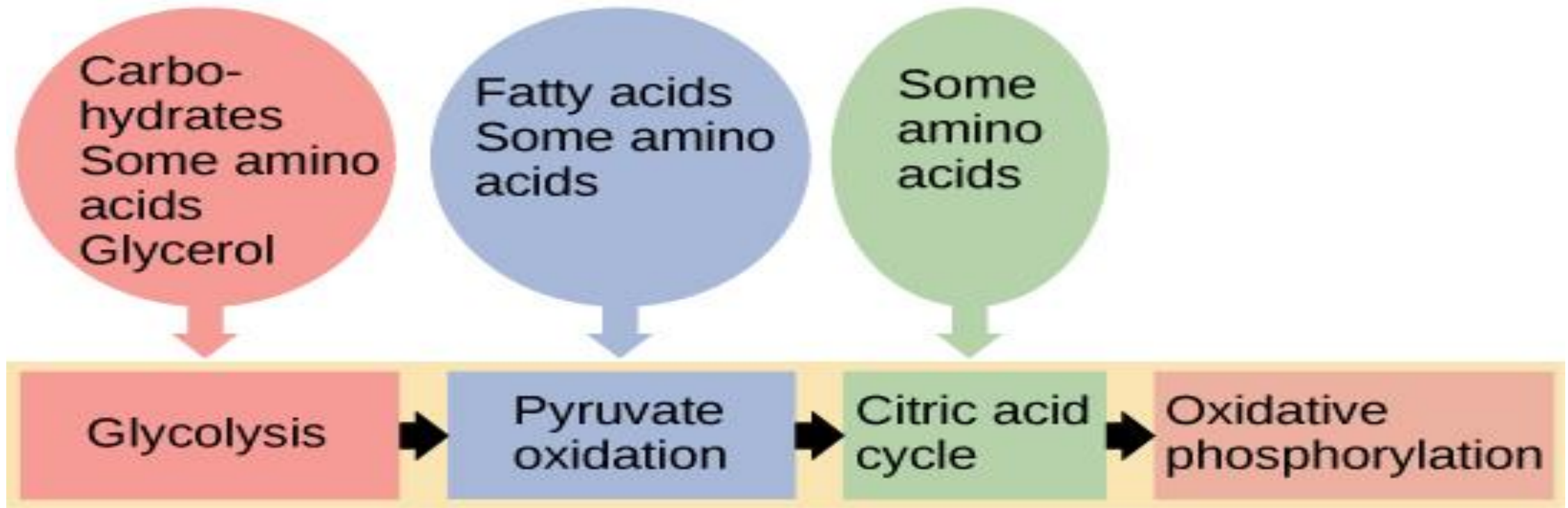


# Anaerobic respiration

- Making energy without the use of O<sub>2</sub>
- Uses inorganic molecules like:
  - Sulfates to make sulfide gas (H<sub>2</sub>S) – bacteria in water wells
  - CO<sub>2</sub> to make methane gas (CH<sub>4</sub>) – stomach ruminants (cows)
- Present in bacteria & archaea

# Macromolecules & energy

- How does a turkey sandwich, which contains protein, provide energy to your cells?
  - This happens because all the catabolic pathways for carbohydrates, proteins, and lipids eventually connect into glycolysis and the citric acid cycle pathways (fig 4.20)
- Carbohydrates – first choice of cellular respiration
- Lipids and Proteins – have different entry points into cellular respiration



- Glycogen from the liver and muscles, together with fats, can feed into the catabolic pathways for carbohydrates.